MANIKYAM MEDIDA

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EDUCATION

B.Tech. in Electronics & Communication Engineering *2022 – 2026*
S R K R Engineering College, Bhimavaram, Andhra Pradesh | **CGPA:** 9.19/10.0

TECHNICAL SKILLS

HDL & Verification: Verilog, SystemVerilog, UVM, OOP, Constrained Randomization, Assertions, Testbench Development, RTL Design, Digital Verification, Functional Coverage, Regression Testing

Protocols: APB, AXI, AHB

Design & Verification Tools: QuestaSim, ModelSim, Cadence Virtuoso, Xilinx Vivado

Programming & Scripting: C, Python, TCL Scripting, Linux

VLSI & Semiconductor: Digital Electronics, ASIC& SoC Basics, FPGA, FSM, DFT, Computer Architecture, CMOS Fabrication, Semiconductor Device Physics

Soft Skills: Analytical problem-solving & debugging, Cross-functional teamwork, Initiative-driven, Strong communication

INTERNSHIPS

VLSI Intern | ElevanceSkills *Sep 2026*
- Developing and simulating an Adaptive Multi-Junction Traffic Signal Controller using VLSI and FSM modelling.
- Applying RTL design, testbench development, digital logic, and simulation techniques to a practical VLSI project.
- Analyzing simulation waveforms and debugging RTL behaviour using ModelSim workflows.

Digital Signal Processing Intern – URSC/ISRO, Bengaluru *Apr 2025*
- Performed transient signal analysis and device characterization using Continuous Wavelet Transform (CWT) for satellite DSP applications.
- Optimized signal processing algorithms with FFT-based convolution, achieving 10× faster computation on large datasets (>14,000 samples at 10 kHz) – directly applicable to test throughput optimization.

VLSI & Semiconductors Intern – IIDT-Blackbucks *May 2025 – Jul 2025*
- Developed & simulated Verilog RTL modules and testbenches, validating digital components through waveform analysis.
- Designed an N-bit Universal Shift Register (parallel load, shift-left, shift-right, hold modes) using flip-flops and 4×1 MUX structures, verifying functional correctness through simulation.

PROJECTS

AXI-to-OCP Interconnect Verification using UVM | SystemVerilog, UVM – QuestaSim *Sep 2026*
- Built a UVM verification environment for an AXI-to-OCP interconnect with two AXI masters and an OCP slave interface.
- Developed a reusable UVM verification environment with agents, sequence items, sequencers, drivers, monitors, scoreboard, and virtual sequences using TLM-based communication and constrained-random stimulus.
- Performed protocol-level verification of AXI handshaking, arbitration, AXI-to-OCP transaction conversion, and response paths, with functional debug using simulation logs and waveform analysis.

APB UVM Verification Environment | SystemVerilog, UVM – QuestaSim *Aug 2026*
- Developed a modular UVM-based APB verification environment with sequencer, driver, monitor, scoreboard, APB Slave BFM, functional coverage, and SystemVerilog assertions.
- Verified constrained-random read/write transactions, wait-state handling, error responses, and protocol assertions.
- Achieved 60/60 scoreboard passes, 0 UVM errors/fatals, and 100% coverage of the defined functional coverage model.

2×2 Ethernet Switch Design & Verification | SystemVerilog – QuestaSim *Jul 2026*
- Designed and verified a 2×2 Ethernet switch, validated destination-based packet routing, SOP/EOP framing, variable packet lengths, and concurrent traffic.
- Built a modular OOP-based verification environment with Packet, Generator, Driver, Monitor, Checker, virtual interfaces, clocking blocks, constrained-randomization, and mailboxes for automated packet verification.
- Added SystemVerilog Assertions to validate packet framing, performed Functional Coverage & Regression Testing.

CERTIFICATIONS & TRAINING

- **VLSI Project Training – ElevanceSkills, 2026:** Practical training in Verilog HDL, RTL Design, FSM modelling, Testbench Development, Digital Logic, and ModelSim/Vivado simulation.
- **Foundation - Design for Testability Certification – Maven Silicon, 2026:** Gained exposure to scan chains, ATPG, boundary scan, fault modeling and VLSI testing fundamentals.
- **ISWDP Samsung Student Fellowship – IISc (with Synopsys & Samsung), 2025:** Semiconductor device physics, CMOS fabrication, and TCAD simulations. Ranked in top 5% of Level 1 performers.
- **Verilog HDL Hands-On Certification – Maven Silicon, 2025:** Practical RTL design and Verilog proficiency.